Dhruv Nirmal

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PROFESSIONAL SUMMARY

I'm a data engineer who builds and automates the pipelines that keep analytics running. I own the end-to-end data pipeline for my firm's largest client, around **6 million** invoice rows refreshed weekly, and I've built and maintained **20+ automated ETL pipelines** across five enterprise clients. I work mainly in Python and advanced SQL, with Databricks and PySpark for distributed processing, cloud on Azure and AWS, and Power BI and Tableau for reporting.

KEY SKILLS

Languages & Pipelines: Python (pandas, numpy), advanced SQL, ETL pipeline design, batch automation, data modelling, data cleaning, feature engineering

Cloud & Orchestration: Azure (Microsoft Graph API), AWS (S3, EC2, IAM, Lambda, Glue, Athena), Windows Task Scheduler, Terraform (IaC), Docker

Big Data & Distributed: Databricks, PySpark, Snowflake, KNIME

Reporting & BI: Power BI, Tableau, dashboard design, KPI reporting

Data Quality: automated validation, anomaly detection, reproducible and documented processes

EXPERIENCE

Data Engineer, Purchasing Index Data Analytics (Comprara Group)

Jun 2024 to Present | Melbourne, Australia

A data and analytics consultancy serving enterprise clients across Australia and New Zealand. I own client data pipelines end to end: ingestion, transformation, validation and delivery. Selected engagements:

Automated ingestion and validation pipeline: largest enterprise client (multi-region)

- Problem: the firm's biggest account ran on a manual KNIME upload-and-check process, around 4 to 5 hours every week, across roughly 6 million invoice rows spanning 5 global regions and 13 sub-regions.
- Approach: replaced it with automated email-based ingestion (Microsoft Graph API into a remote SQL Server via batch scripts), and built automated validation (row and hour counts, data-type and column-consistency checks) that drops stray columns, logs discrepancies and emails client and internal teams. Downstream I run vendor and spend lookups, multi-language cleaning, an ~11 to 12 thousand-rule categorisation, and rolling 4-week spend windows into dashboards. Python, advanced SQL (SQL Server), batch automation, Azure.
- Result: cut weekly processing from ~4 to 5 hours to a ~75-minute automated run, around 3 to 3.5 hours saved every week, and standardised intake so all 13 regions feed one consistent pipeline.

ETL pipeline platform: five enterprise clients

- Problem: reporting across millions of procurement transactions each month for five clients was manual and slow.
- Approach: built and maintained 20+ automated ETL pipelines (Python, SQL, batch files) deployed on Azure-hosted servers with Windows Task Scheduler.
- Result: reduced manual reporting effort by around 40% and improved turnaround by around 50%, with production-grade reliability.

Categorisation pipeline: ~A$12B enterprise spend

- Problem: categorising a client's five-year, roughly A$12B spend book was manual and took an account manager 1 to 1.5 months per cycle.
- Approach: built a documented, repeatable Python pipeline (scikit-learn, TF-IDF, one-vs-all classifier) with feature engineering on spend value and line count, and productionised it for reuse.
- Result: cut the cycle to a single day's run and rolled the pipeline out across other client accounts.

Reporting and BI: built and maintained 16+ Tableau dashboards and a Power BI solution for procurement and finance stakeholders, giving live visibility into spend, supplier performance and KPIs.

Data Engineer Intern, Victorian Centre for Data Insights (VCDI)

Aug 2023 to Nov 2023 | Melbourne, Australia

Distributed anomaly-detection pipeline (Databricks, PySpark)

- Problem: government procurement data was too large to screen on a single machine.
- Approach: built a distributed anomaly-detection pipeline in Databricks with PySpark, engineering scalable transformation layers, and delivered the findings through a Power BI solution.
- Result: improved detection accuracy by about 20%; the Power BI solution was adopted by senior Department of Transport stakeholders.

PROJECTS

Cloud Data Platform (Terraform, AWS, Snowflake) -- personal build

- Provisioned full cloud infrastructure as code with Terraform (AWS S3, IAM, EC2, networking), built an automated ingestion pipeline from external APIs into Snowflake, and used Docker for a modular, version-controlled multi-environment setup (dev / staging / prod).

Facit: multi-source data integration -- personal build

- Built a pipeline unifying POS, wages and 14 suppliers' invoices into one model behind a weekly analytics product for independent cafés. Live with a St Kilda café as the design partner.

EDUCATION

Master of Data Science, Monash University | Feb 2022 to Dec 2023

Bachelor of Engineering, Thapar University | May 2017 to May 2021